

1. Administrative & Study Identification

Full Study Title: Safety and Efficacy Study of Enzalutamide in Patients With Nonmetastatic Castration-Resistant Prostate Cancer
Short Title/Acronym: PROSPER **Protocol Number:** MDV3100-14 **NCT Number:** NCT02003924 **Trial Status:** Active but not currently recruiting (ACTIVE_NOT_RECRUITING) **Sponsor Name:** Pfizer / Astellas Pharma **Investigational Product:** Enzalutamide (Xtandi) **ATC Code:** L02BB04 **Phase of Development:** Phase III **Medical Monitor:** Clinical Operations Lead, Pfizer / Astellas Pharma

2. Study Synopsis & Rationale

2.1 Study Objective

Primary Objective: To evaluate the efficacy of enzalutamide in extending Metastasis-Free Survival (MFS) compared to placebo in patients with nonmetastatic castration-resistant prostate cancer (nmCRPC) who are continuing androgen deprivation therapy (ADT).
Secondary Objectives: To evaluate overall survival (OS), time to prostate-specific antigen (PSA) progression, time to first use of new antineoplastic therapy, time to first use of cytotoxic chemotherapy, chemotherapy-free disease-specific survival (CFDS), chemotherapy-free survival (CFS), and overall safety/tolerability.

2.2 Background & Rationale To address the lack of standard care to delay metastasis in men with nonmetastatic castration-resistant prostate cancer (nmCRPC) who have a rapidly rising PSA. Men with nmCRPC and a PSA doubling time ≤ 10 months are at high risk for developing metastatic disease, which is associated with significant morbidity and mortality. Enzalutamide is an oral androgen receptor inhibitor that targets multiple steps in the androgen receptor signaling pathway, and this study assesses its ability to delay the onset of metastasis and improve overall survival.

3. Study Design & Methodology

3.1 Design Framework

Study Type: Interventional **Control Type:** Placebo-controlled
Blinding: Double-blind **Randomization:** 2:1 (Enzalutamide : Placebo), stratified by PSA doubling time (< 6 months vs. 6-10 months) and baseline use of bone-targeting agents (yes vs. no).

3.2 Planned Enrollment & Duration

Target N\$: 1,401 **Number of Sites:** > 200 sites globally **Estimated Study Start:** December 2013 **Estimated Primary Completion:** June 2019 (Final OS data cutoff: October 2019)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Histologically or cytologically confirmed diagnosis of prostate adenocarcinoma without neuroendocrine differentiation, signet cell, or small-cell features. 2. Asymptomatic nonmetastatic castration-resistant prostate cancer (nmCRPC) with a PSA doubling time ≤ 10 months and screening PSA assessed by central laboratory ≥ 2 $\mu\text{g/L}$ (2 ng/mL). 3. Ongoing androgen deprivation therapy (ADT) with a gonadotropin-releasing hormone (GnRH) agonist/antagonist or prior bilateral orchiectomy, with castrate levels of serum testosterone ≤ 50 ng/dL (≤ 1.73 nmol/L) at screening. 4. Progressive disease on androgen deprivation therapy at enrollment. 5. Eastern Cooperative Oncology Group (ECOG) performance status of 0 or 1. 6. Estimated life expectancy ≥ 12 months. 7. Adequate hematologic, hepatic, and renal function.

4.2 Exclusion Criteria 1. Prior or present evidence of metastatic disease, assessed by whole-body radionuclide bone scan and CT/MRI. 2. Prior cytotoxic chemotherapy for prostate cancer. 3. Use of hormonal therapy or biologic therapy for prostate cancer (other than approved bone targeting agents and GnRH agonist/antagonist therapy) or use of an investigational agent within 4 weeks of randomization. 4. Known or suspected brain metastasis or active leptomeningeal disease. 5. History of another invasive cancer within 3 years of randomization. 6. Absolute neutrophil count $< 1000/\mu\text{L}$, platelet count $< 100,000/\mu\text{L}$, or hemoglobin < 10 g/dL (6.2 mmol/L) at screening. 7. Total bilirubin ≥ 1.5 times the upper limit of normal; Creatinine > 2 mg/dL (177 $\mu\text{mol/L}$); or Albumin < 3.0 g/dL (30 g/L) at screening. 8. History of seizure or any condition that may predispose to seizure. 9. Clinically significant cardiovascular disease or gastrointestinal disorder affecting absorption. 10. Major surgery within 4 weeks of randomization. 11. Hypersensitivity reaction to the active pharmaceutical ingredient or any of the capsule components, including Labrasol, butylated hydroxyanisole, and butylated hydroxytoluene. 12. Any severe concurrent disease, infection, or comorbid condition that would interfere with study participation, place the patient at undue risk, or complicate interpretation of data.

5. Intervention & Treatment Plan

5.1 Investigational Regimen

Dosage/Frequency: Enzalutamide 160 mg/day (administered as four 40-mg soft gelatin capsules) or matching placebo once daily.
Route of Administration: Oral **Duration of Treatment:** Continued until radiographic progression, unacceptable toxicity, or withdrawal of consent.

5.2 Concomitant & Prohibited Therapy

Permitted: Background Androgen Deprivation Therapy (ADT) using a gonadotropin-releasing hormone agonist/antagonist must be maintained. Bone-targeting agents are permitted. **Prohibited:** Concurrent use of other investigational antineoplastic agents, cytotoxic chemotherapy, or strong CYP3A4/CYP2C8 inhibitors/inducers where alternative medication is available.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, PSA verification, Baseline Imaging (CT/MRI/Bone Scan)
Baseline	Day 0	Randomization, First Dose, QoL Baseline
Interim	Every 16 Weeks	Onsite Visits: Imaging (CT/MRI/Bone Scan) for MFS assessment, Safety Labs, PSA tests, FACT-P QoL
End of Study	Variable	Treatment Discontinuation visit (radiographic progression), Survival and subsequent therapy follow-up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Metastasis-Free Survival (MFS), defined as the time from randomization to first date of radiographic progression (RP) by blinded independent central radiology review at any time, or death within 112 days of treatment discontinuation without evidence of RP. **Measurement Method:** Central independent blinded radiographic review of whole-body radionuclide bone scans for bone disease (appearance of ≥ 1 metastatic lesions) and CT/MRI for soft tissue disease (per RECIST 1.1). Participants who did not have an MFS event at the time of analysis data cut-off (28 June 2017) were censored at the date of last assessment showing no objective evidence of RP prior to skeletal-related event or two or more consecutive missed tumor assessments.

7.2 Secondary & Exploratory Endpoints * Overall Survival (OS) * Time to prostate-specific antigen (PSA) progression * Time to first use of new antineoplastic therapy * Time to first use of cytotoxic chemotherapy * Quality of life measures (FACT-P global score, EQ-5D-5L)

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard continuous AE collection; specific focus on events of special interest for AR inhibitors, including falls, fatigue, hypertension, and cardiovascular events. **Serious Adverse Events (SAEs):** Must be reported to the sponsor within 24 hours of site awareness. **Data Monitoring Committee (DMC):** An Independent Data Monitoring Committee (IDMC) reviewed safety and efficacy data periodically throughout the trial.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for all efficacy endpoints. Safety Set for all patients who received at least one dose of the study drug. **Sample Size Justification:** A minimum of 440 MFS events provided 90% power to detect a target hazard ratio of 0.72 using a 2-sided log-rank test at an overall significance level of $\alpha = 0.05$. **Handling of Missing Data:** Analysis was based on Kaplan-Meier estimates. Patients without an MFS event were censored at the time of their last adequate tumor assessment. Time-to-event endpoints utilized stratified Cox proportional hazards models.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular clinical onsite and remote monitoring to ensure protocol adherence. **Audit Trail:** Secure eCRF systems, robust data clarification form (DCF) processes, and central blinded imaging review to ensure objective primary endpoint evaluation.

11. Study Identifiers & Governance

Posting ID: 217472311191024771 **Primary ID:** MDV3100-14 **Registry Number:** NCT02003924 **Sponsor/Lead Organization:** Pfizer / Astellas Pharma **Study Title:** PROSPER **Official Regulatory Title:** PROSPER: A MULTINATIONAL, PHASE 3, RANDOMIZED, DOUBLE-BLIND, PLACEBO-CONTROLLED, EFFICACY AND SAFETY STUDY OF ENZALUTAMIDE IN PATIENTS WITH NONMETASTATIC CASTRATION-RESISTANT PROSTATE CANCER

12. Clinical Framework (Prostate Cancer Specific)

Disease Areas: Nonmetastatic Castration-Resistant Prostate Cancer, Prostate Cancer, Cancer of the Prostate **Preferred UMLS Name:** Nonmetastatic Castration-Resistant Prostate Carcinoma

Primary Condition: Nonmetastatic Castration-Resistant Prostate Cancer (nmCRPC) **Diagnosis Threshold:** PSA doubling time ≤ 10 months and PSA ≥ 2 ng/mL despite castrate testosterone **Medical Methodology:** Androgen Receptor Inhibition combined with continuous Androgen Deprivation Therapy (ADT) **Optimization Target:** Delay of distant metastasis and prolonged Overall Survival (OS)

13. Study Procedures & Methodology

Management Pathway: Standard oncology clinical pathway for CRPC continuous management **Education Protocol (HCP):** Protocol training, safety/AE monitoring (especially falls and fatigue), and imaging requirements **Education Protocol (Patient):** Oral drug administration instructions, adherence tracking, and AE reporting **Quality Audit Mechanism:** Central independent blinded radiographic review for MFS

14. Observation & Follow-Up Schedule

Total Duration: Continued until death, withdrawal, or study termination (median OS follow-up ~48 months) **Onsite Visit Cadence:** Every 16 weeks for imaging and efficacy assessments **Remote Monitoring Frequency:** Routine pharmacovigilance and survival follow-ups post-discontinuation **Checklist Review Interval:** Quality of Life assessments (FACT-P) administered at scheduled clinic visits

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				